

Problem F. Tower Platform

TimeLimit: 1 second
MemoryLimit: 1024 megabytes

In the distant Sky City, there is a path consisting of n high towers. These towers are arranged from left to right, where the position of the i -th tower is (i, h_i) , and h_i represents the height of the i -th tower.

The explorer EZ must travel from the top of the 1st tower to the top of the n -th tower. Since there are no roads between the towers, without any special facilities, he can only move from the top of the i -th tower to the top of the adjacent $(i + 1)$ -th tower. The distance required to move from the top of the i -th tower to the top of the $(i + 1)$ -th tower is defined as the Euclidean distance between the tops of the two towers:

$$\sqrt{(i - (i + 1))^2 + (h_i - h_{i+1})^2}.$$

EZ has obtained **exactly one** special sky platform. If he chooses to build the platform between the i -th and j -th towers, the platform will connect the tops of the two towers in a straight line. EZ can then travel along this platform directly from the top of the i -th tower to the top of the j -th tower without passing through the intermediate towers. The distance required to travel along the platform to the j -th tower is the length of the platform, which is the Euclidean distance between the tops of the two towers:

$$\sqrt{(i - j)^2 + (h_i - h_j)^2}.$$

However, the platform must satisfy the following condition:

The interior of the platform cannot pass through or touch any other towers.

In other words, for all towers k satisfying $i < k < j$, the top of the k -th tower must lie below the platform. Please determine between which two towers the platform should be built to minimize the total distance EZ needs to travel from the top of the 1-st tower to the top of the n -th tower.

Input

The first line contains a single integer n — denotes the number of towers.

The second line contains n integers $h_1, h_2, \dots, h_n$ — denotes height of each tower.

- $2 \leq n \leq 10^6$
- $1 \leq h_i \leq 10^9$

Output

Output a real number representing the minimum total distance required for the explorer EZ to travel from the 1-st tower to the n -th tower under the optimal platform construction.

Your answer will be considered correct if its absolute or relative error does not exceed 10^{-6} . Formally, let your answer be a , the jury's answer be b . Your answer will be considered correct if $\frac{|a-b|}{\max(1, |b|)} \leq 10^{-6}$.

Examples

standard input	standard output
5 1 4 2 5 3	7.63441362
2 1 1000000000	999999999.00000000